AJ SMART BIZ · DOCUMENT 07

Deployment, DevOps & Operations Runbook

Local setup, environments, Docker images, CI/CD pipelines, nginx and DNS, and day-2 operations.

Audience	DevOps, developers, on-call
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Sources	Dockerfiles, `.github/workflows/*` (at HEAD), `deploy/`, branch `nginx-domain-live`

Contents

1. Local development
2. Configuration (environment variables)
3. Production topology
4. Docker images
5. CI/CD pipelines
6. nginx, DNS and TLS
7. Onboarding a tenant domain
8. Operations runbook

1. Local development

App	Folder	Command	URL
API	Aj-Smart-Biz-Backend	`npm install && npm run dev`	http://localhost:4000/api/v1 (docs at /api/v1/docs)
Super Admin	Aj-Smart-Biz-Supper-Admin	`npm install && npm start`	http://localhost:4200
Company Admin	Aj-Smart-Biz-Admin	`npm install && npm start`	http://localhost:4300
Theme default / midnight / atelier	themes/<name>	`npm run dev`	:4400 / :4500 / :4600 — use `salon.localhost:4500` or `?domain=`
Product website	Aj-Smart-Biz-Technology	`npm run dev`	http://localhost:4700

- MySQL must be running; the API creates its database. No MySQL? `DB_DIALECT=sqlite npm run dev`.
- First boot seeds the root super admin from `.env.development` (default `superadmin@ajsmartbiz.com`) and prints the credentials — change before any deployment.
- Angular dev servers proxy `/api/v1` and `/uploads` to `API_PROXY_TARGET` (default `http://localhost:4000`) via `proxy.conf.js`.
- Demo data (opt-in, idempotent): `npm run db:seed:salon`, `db:seed:decor`, `db:seed:catalogue [companyId]`, `db:seed:blog`, `db:seed:marketing`.
- After testing a Next.js site with `next dev`, run `npm run build` again before `npm start` — the dev server overwrites the production build.

2. Configuration

Backend (`.env.development` / `.env.production`; real env wins)

Variable	Purpose
NODE_ENV, PORT, API_PREFIX	Mode, port (4000), prefix (/api/v1)
DB_DIALECT, DB_HOST, DB_PORT, DB_NAME, DB_USER, DB_PASSWORD, DB_STORAGE, DB_LOGGING	Database
DB_SYNC_MODE	alter (default) · force (refused in production) · none
JWT_SECRET, JWT_EXPIRES_IN, JWT_REFRESH_SECRET, JWT_REFRESH_EXPIRES_IN	Tokens (boot refuses placeholder secrets in production)
SUPER_ADMIN_NAME/EMAIL/PHONE/PASSWORD	Root account ensured on every boot

Variable	Purpose
CORS_ORIGINS	Comma list; `*` reflects origin (dev tunnels only)
BCRYPT_SALT_ROUNDS, DEFAULT_COMPANY_ADMIN_PASSWORD	Hashing cost; fallback main-admin password
UPLOAD_DIR, UPLOAD_PUBLIC_PATH, UPLOAD_MAX_SIZE_MB	Uploads (default 2 MB per file)
CUSTOMER_OTP_DEV_CODE	Fixed OTP until an SMS gateway exists — not production-safe
ANTHROPIC_API_KEY, AI_CONTENT_MODEL	AI content creator

Websites

Variable	App	Purpose
API_URL	both	Server-side API base, e.g. `http://aj-smart-biz-backend:4000/api/v1` (no trailing slash)
SITE_URL, SITE_SNAPSHOT_DIR	Technology	Canonical URL; where the last good bootstrap snapshot is kept
TENANT_DOMAIN	themes	Dev fallback only — leave unset in production
NEXT_PUBLIC_FILES_URL	themes (build-time)	CDN origin for uploads
UPLOADS_ORIGIN	themes	Target of the /uploads proxy (defaults to API_URL origin)
ALLOWED_ORIGINS	both	Server Action origins for dev tunnels; empty in production
TZ, HOST_PORT	deploy env files	Container timezone; published host port

Angular consoles

`src/environments/environment.ts` (production) holds `apiUrl` and `filesUrl`; development uses relative `/api/v1` through the proxy. The nginx image can also proxy `/api/v1` to `API_ORIGIN` (default `http://aj-smart-biz-backend:4000`).

3. Production topology

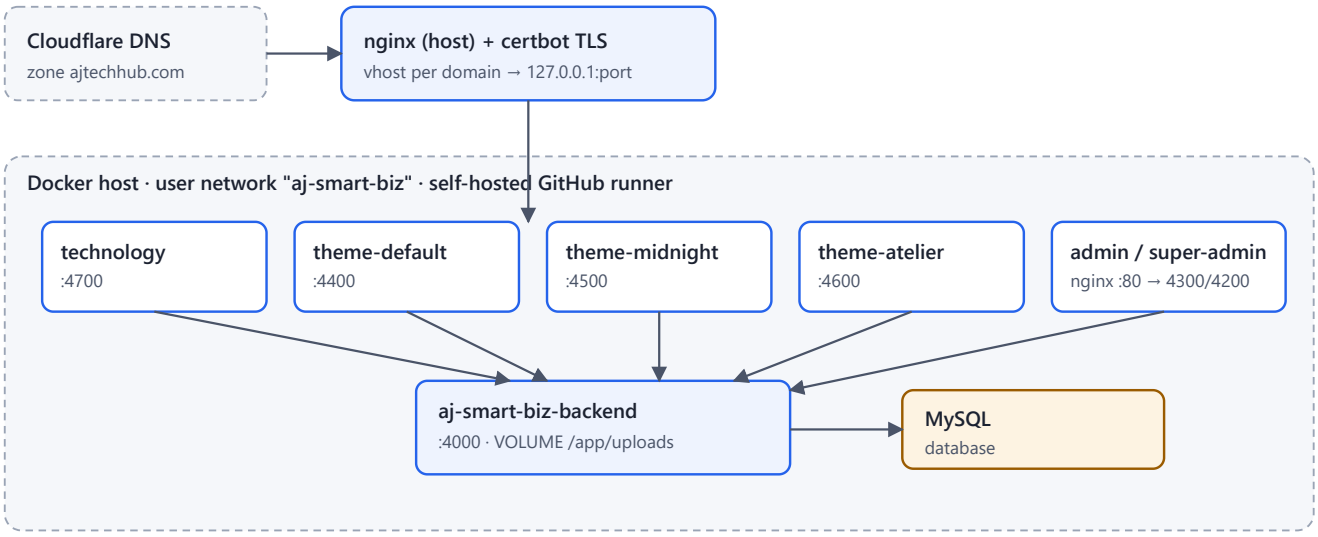


Figure 1 — One server: nginx terminates TLS and proxies each domain to a container port; containers reach the API by name on the Docker network.

4. Docker images

Image	Build	Runtime	Health check
Backend	node:24-alpine, `npm ci --omit=dev`	user node, port 4000, `VOLUME /app/uploads`	GET /
Super Admin / Admin	node:24-alpine, `ng build --configuration production` (needs internet for font inlining)	nginx:1.29-alpine on :80; envsubst `API_ORIGIN`; gzip; hashed assets cached 1 year; SPA fallback; `/api/v1` and `/uploads` proxied (25 MB body, 120 s timeout)	wget /index.html
Themes / Technology	3 stages deps → build → runtime, node:24-alpine + libc6-compat; themes take build arg `NEXT_PUBLIC_FILES_URL`	Next standalone `node server.js` as user node, HOSTNAME 0.0.0.0, telemetry off	GET / every 30 s

Registry: Docker Hub repository `ajtechhub/ajtechhub` with service-prefixed tags.

5. CI/CD pipelines

Each service has two GitHub Actions workflows triggered by pushing to a dedicated branch.

Workflow	Trigger	What it does
`<service>-live.yml`	push to `<service>-live` or manual	Build with buildx and push only. Tags `<service>-1.0.<run>` and `<service>-live`.
`<service>-deploy.yml`	push to `<service>-deploy`	Job 1 (ubuntu-latest): build, push `<service>-deploy-1.0.<run>`, output digest. Job 2 (self-hosted runner on the server): validate `/opt/aj-smart-biz/<svc>.env` (API_URL set, unquoted, no trailing slash, no CRLF), pull by digest, `docker run -d --restart unless-stopped --network aj-smart-biz --env-file … -p HOST_PORT:APP_PORT`, wait ≤ 2 min for healthy, **roll back** to the previous image on failure, prune.
`docker-publish.yml`	push to `dev_live` touching backend	Backend image tagged `aj-smart-biz-backend-dev_live`.
`deploy-nginx-domains.yml`	push to `nginx-domain-live`	Cloudflare A records → sync `domains/*.conf` to `/etc/nginx/conf.d/managed-*.conf` → `nginx -t` + reload (rollback on failure) → certbot `--nginx`.

Services: aj-smart-biz-backend, -admin, -supper-admin, -technology, -theme-default, -theme-midnight, -theme-atelier.

Secrets/vars: DOCKERHUB_USERNAME/TOKEN, <SERVICE>_ENV_FILE, <SERVICE>_HOST_PORT, theme build args (WHITE_THEME_*, BLACK_THEME_*, DECOR_FRAMING_*), NGINX_SERVER_*, CLOUDFLARE_API_TOKEN/ZONE_ID/ZONE.

> **Working tree note**
>
> The workflow files are deleted in the current working tree of branch `aj-smart-biz-backend-deploy` (uncommitted). They exist at HEAD. Confirm whether this deletion is intended before committing.

6. nginx, DNS and TLS

Domain	Upstream	Body limit
smart-biz.ajtechhub.com	4700 Technology	2 MB
smart-biz-white.ajtechhub.com	4400 theme default	8 MB
smart-biz-black.ajtechhub.com	4500 theme midnight	8 MB
smart-biz-decor-framing.ajtechhub.com	4600 theme atelier	8 MB
smart-biz-api.ajtechhub.com	4000 API	25 MB
smart-biz-admin.ajtechhub.com	4300 Company Admin	4 MB

Domain	Upstream	Body limit
smart-biz-super-admin.ajtechhub.com	4200 Super Admin	4 MB

Vhosts listen on 80 (certbot adds 443 and the redirect) and forward `Host`, `X-Real-IP`, `X-Forwarded-For`, `X-Forwarded-Proto`, `X-Forwarded-Host` and WebSocket upgrade headers.

> **Check the API hostname**
> The Angular production environments point at `aj-smart-biz-api.ajtechhub.com`, while the nginx vhost is `smart-biz-api.ajtechhub.com`. Confirm which one is live and align them.

7. Onboarding a tenant domain

1. In the Super Admin (company → Domains) or Company Admin (Company Details → Domains), add the bare host, optionally pinned to a branch.
2. Add a vhost in `nginx-domains/domains/<host>.conf` on branch `nginx-domain-live`, proxying to the chosen design's port (4400/4500/4600). Push — DNS, nginx and TLS are applied by the workflow.
3. For domains not in the Cloudflare zone, the tenant points an A record to the server IP first.
4. Open the site; the middleware resolves the tenant from `X-Forwarded-Host`.

8. Operations runbook

Situation	Action
Website shows "Service unavailable"	API unreachable from the container: check `docker ps`, API health (`/api/v1/health`), network `aj-smart-biz`, `API_URL` in the env file.
Website shows "Company not found"	Domain row missing/inactive, company inactive, or vhost not forwarding `X-Forwarded-Host`.
Website shows holding page	No running subscription — check Company Plans; reactivate or assign a plan.
Plan did not expire/start on time	Super Admin → Company Plans → *Run expiry check* (`POST /super-admin/subscriptions/run-due`); check API logs for the hourly sweeper.
Deploy failed	Workflow rolls back automatically; read job 2 logs (env file validation, health timeout).
Uploads missing after redeploy	Ensure the backend container mounts a persistent volume at `/app/uploads`.
Too many indexes (MySQL ER_TOO_MANY_KEYS)	Boot drops duplicate `key_N` indexes before alter-sync; restart the API.
Backups	Schedule `mysqldump` of the database and an archive of the uploads volume; take one before any model change is deployed.

Situation	Action
Rotate secrets	Change JWT secrets (signs everyone out), super-admin password, Docker Hub token; update server env files.